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$$I = \int \frac{2x+7}{(x^2+1)^2} dx = \int \frac{2x}{(x^2+1)^2} dx + 7 \int \frac{1}{(x^2+1)^2} dx$$

$$\text{Stel: } t = x^2 + 1 \Rightarrow dt = 2x dx$$

$$\begin{aligned} &= \int \frac{1}{t^2} dt + 7 \int \frac{1}{(x^2+1)^2} dx \\ &= \frac{1}{-1t} + c + 7 \left[\frac{2 \cdot 2 - 3}{2 \cdot 2 - 2} \cdot \int \frac{1}{(x^2+1)} dx + \frac{x}{(2 \cdot 2 - 2)(x^2+1)} \right] \\ &= \frac{-1}{x^2+1} + c + 7 \left[\frac{1}{2} \cdot \text{Bgtan}(x) + c + \frac{x}{2(x^2+1)} \right] \\ &= \frac{-1}{x^2+1} + \frac{7}{2} \text{Bgtan}(x) + \frac{7x}{2(x^2+1)} + c \\ &= \frac{-2+7x}{2(x^2+1)} + \frac{7}{2} \text{Bgtan}(x) + c \end{aligned}$$

References